Model Predictive Control

Mini final project

by

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Introduction

Model Predictive Control (MPC) is a powerful tool for managing complex systems with the constraints included in the design procedure, making it ideal for advanced vehicle control applications. This project focuses on designing different MPC cruise controllers for a car driving on a highway, with the aim of ensuring safety, efficiency, and passenger comfort under dynamic conditions.

Driving on the highway involves challenges such as strict operational constraints and the need to respond to varying scenarios such as maintaining target speed, lane changing, and overtaking. This project addresses these issues by developing linear and nonlinear MPC controllers capable of robust performance while satisfying constraints on states and inputs.

1.1. Overview

This project comprises five main tasks, described in Deliverables 2 through 6, each addressed in a separate section.

Deliverable 2 discusses the linearization of the system model, as the linearized model is used for control in Deliverables 3 through 5. Deliverable 2.1 derives the analytical expression for this linearized model, while Deliverable 2.2 provides an intuitive mechanical explanation of the feasibility of decomposing the system into two independent longitudinal and lateral subsystems. It is important to note that these continuous models are subsequently discretized, as the MPC controllers implemented in later sections are discrete.

Deliverable 3 involves the design of two recursively feasible, stabilizing MPC controllers that track references for each subsystem. These controllers ensure recursive satisfaction of the problem's constraints. This section also includes plotting the terminal invariant set for the lateral subsystem.

Deliverable 4 enhances the previously implemented controller to reject disturbances and track setpoint references without offset (offset-free tracking). These disturbances arise due to the linearization errors introduced by the motor and drag terms in the system dynamics.

Deliverable 5 focuses on designing a robust tube MPC controller for adaptive cruise control, enabling the tracking of a reference velocity while maintaining a minimum distance from a lead vehicle exhibiting uncertain behavior (i.e., changing velocity unpredictably). This section also includes plotting the minimal invariant set, denoted as ε, and the terminal set, denoted as χ_f.

Deliverable 6 develops nonlinear MPC (NMPC) controllers that satisfy the problem's constraints while operating on the nonlinear system model. This approach eliminates the need for subsystem decomposition and covers the entire state space. Deliverable 6.1 presents the design of an NMPC controller for tracking a desired reference, while Deliverable 6.2 focuses on an NMPC controller for performing an overtaking maneuver. The overtaking is achieved through the incorporation of an ellipsoidal collision avoidance constraint.

1.2. System Model

A four-state kinematic model of a car moving in a 2D plane is considered the system. The state vector is defined as equation 1.1.

$$\mathbf{x} = \begin{bmatrix} x \\ y \\ \theta \\ V \end{bmatrix}^T, \quad [\mathbf{x}] = \begin{bmatrix} \text{m} \\ \text{m} \\ \text{rad} \\ \text{m/s} \end{bmatrix}^T \tag{1.1}$$

where (x, y) represents the position of the car's center of mass in the world frame, θ is the heading angle of the car with respect to the x-axis, and V is the velocity of the vehicle. The input vector of the model is defined as equation 1.2.

$$\mathbf{u} = \begin{bmatrix} \delta \\ u_T \end{bmatrix}^T, \quad [\mathbf{u}] = \begin{bmatrix} \text{rad} \\ \text{—} \end{bmatrix}^T \tag{1.2}$$

where δ is the steering angle of the front wheels, and u_T is the normalized throttle to the motor.

1.2.1. System Kinematics and Dynamics

The motion of the car is captured with a bicycle model. Slip angle β is the angle between the actual direction of travel of the car and the direction in which it is pointed. This angle is defined at the center of mass of the car and is computed based on the steering geometry as equation 1.3.

$$\beta = \arctan\left(\frac{\ell_r \tan(\delta)}{\ell_r + \ell_f}\right) \tag{1.3}$$

where ℓ_r and ℓ_f are the distances from the center of mass to the rear and front axles, respectively.
The longitudinal dynamics are influenced by the motor force, aerodynamic drag, and roll resistance as expressed in equations 1.4 to 1.6, respectively.

$$F_{\text{motor}} = \frac{u_T P_{\text{max}}}{V} \tag{1.4}$$

$$F_{\text{drag}} = \frac{1}{2}\rho C_d A_f V^2 \tag{1.5}$$

$$F_{\text{roll}} = C_r m g \tag{1.6}$$

where m is the vehicle mass, P_{max} is the maximum motor power, ρ is the air density, C_d is the drag coefficient, A_f is the frontal area of the vehicle, C_r is the roll coefficient, and g is the gravity constant. This model is only valid for $V > 0$, which is hold on the highway drive.
In equation 1.4, a positive value for u_T represents supplying the car with power from the motor, a negative value for this variable represents braking, and $u_T = 0$ represents no motor force.
The complete state equations are expressed in equation 1.7

$$\dot{\mathbf{x}} = f(\mathbf{x}, \mathbf{u}) = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{V} \end{bmatrix} = \begin{bmatrix} V\cos(\theta + \beta) \\ V\sin(\theta + \beta) \\ \frac{V}{\ell_r}\sin(\beta) \\ \frac{F_{\text{motor}} - F_{\text{drag}} - F_{\text{roll}}}{m} \end{bmatrix} \tag{1.7}$$

1.2.2. System Constraints

Since the car drives at highway speeds, it should always stay in the lanes, which imposes a limit on the state y based on 1.2. To increase passengers' comfort, maximum heading angle of the car during a lane change is limited. Apart from comfort, this limitation also nicely bounds the linearization error. These two constraints are expressed in equations 1.8 and 1.9 respectively.

$$-0.5\,\text{m} \leq y \leq 3.5\,\text{m} \tag{1.8}$$

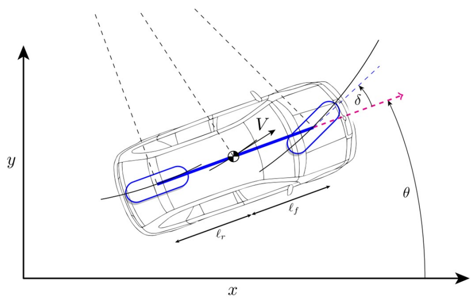


Figure 1.1: Car model

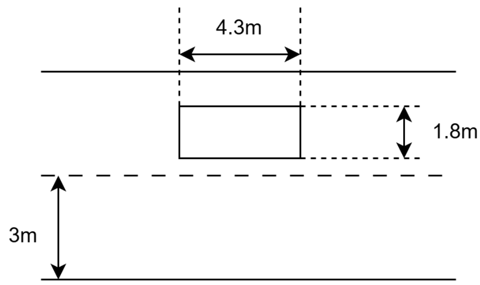


Figure 1.2: Problem Setup

$$|\theta| \leq 5^\circ = 0.0873\,\text{rad} \tag{1.9}$$

Note: Since the optimal operating point for lateral controllers in different sections are at the limit of the constraint $\theta = 5°$, $|\theta| <= 0.0872$ is considered in the matlab files to stay strictly under $5°$.
The inputs are constrained by mechanical limitations, expressed by equations 1.10 and 1.11.

$$-1 \leq u_T \leq 1 \tag{1.10}$$

$$|\delta| \leq 30^\circ = 0.5236\,\text{rad} \tag{1.11}$$

2

MPC

2.1. Linearization (Deliverable 2.1)

In this section, linearization is discussed. This linearization is being done in the x-direction of travel, i.e., the longitudinal motion. Since the objective is cruise control for a target velocity reference, there exists no steady state to linearize the system around (the x position is integrated in time with a positive velocity). Thus, $f(\mathbf{x}_s, \mathbf{u}_s) \neq 0$. The steady state of the subsystem is considered as equation 2.1.

$$f(\mathbf{x}_s, \mathbf{u}_s) = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{V} \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \\ 0 \\ 0 \end{bmatrix}, \tag{2.1}$$

The analytical expression of $f(\mathbf{x}_s, \mathbf{u}_s)$ is derived by substituting $\mathbf{x}_s$ and $\mathbf{u}_s$ in the system dynamics $f(\mathbf{x}, \mathbf{u})$ and is mentioned as followed. The linearized dynamics are found through a Taylor series expansion up to first order of equation (2.2)

$$f(\mathbf{x}, \mathbf{u}) \approx f(\mathbf{x}_s, \mathbf{u}_s) + A(\mathbf{x} - \mathbf{x}_s) + B(\mathbf{u} - \mathbf{u}_s), \tag{2.2}$$

$$\text{where } A = \left.\frac{\partial f(\mathbf{x}, \mathbf{u})}{\partial \mathbf{x}}\right|_{(\mathbf{x}_s, \mathbf{u}_s)}, \tag{2.3}$$

$$\text{and} \quad B = \left.\frac{\partial f(\mathbf{x}, \mathbf{u})}{\partial \mathbf{u}}\right|_{(\mathbf{x}_s, \mathbf{u}_s)}. \tag{2.4}$$

Calculation of the derivatives expressed in equation (2.3) and equation (2.4) results in equation (2.5) and equation (2.6) respectively.

$$\frac{\partial f(\mathbf{x}, \mathbf{u})}{\partial \mathbf{x}} = \begin{bmatrix} 0 & 0 & -V sin(\theta+\beta) & cos(\theta+\beta) \\ 0 & 0 & V cos(\theta+\beta) & sin(\theta+\beta) \\ 0 & 0 & 0 & \frac{sin(\beta)}{\ell_r} \\ 0 & 0 & 0 & -\frac{u_T P_{max}}{V^2 m} - \frac{\rho C_d A_f V}{m} \end{bmatrix} \tag{2.5}$$

$$\frac{\partial f(\mathbf{x}, \mathbf{u})}{\partial \mathbf{u}} = \begin{bmatrix} -V sin(\theta+\beta)\frac{d\beta}{d\delta} & 0 \\ V cos(\theta+\beta)\frac{d\beta}{d\delta} & 0 \\ \frac{V}{\ell_r} cos(\beta)\frac{d\beta}{d\delta} & 0 \\ 0 & \frac{P_{max}}{Vm} \end{bmatrix} \tag{2.6}$$

where

$$\frac{d\beta}{d\delta} = \frac{\ell_r(\ell_r + \ell_f)}{(\ell_r^2 tan^2(\delta) + (\ell_r + \ell_f)^2) cos^2(\delta)} \tag{2.7}$$

By substituting $\mathbf{x}_s = (0, 0, 0, V_s)^T$ and $\mathbf{u}_s = (0, u_{T,s})^T$ in the equation (2.2), equation (2.5), and equation (2.6) the expected analytical expressions of deliverable 2.1 are obtained.

$$f(\mathbf{x}_s, \mathbf{u}_s) = \begin{bmatrix} V_s \\ 0 \\ 0 \\ \frac{u_{T,s} P_{max}}{m V_s} - \frac{\rho C_d A_f V_s^2}{2m} - C_r g \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \\ 0 \\ 0 \end{bmatrix}, \tag{2.8}$$

$$A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & V_s & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -\frac{U_{T,s} P_{max}}{V_s^2 m} - \frac{\rho C_d A_f V_s}{m} \end{bmatrix} \tag{2.9}$$

$$B = \begin{bmatrix} 0 & 0 \\ V_s \frac{\ell_r}{\ell_r + \ell_f} & 0 \\ \frac{V_s}{\ell_r + \ell_f} & 0 \\ 0 & \frac{P_{max}}{V_s m} \end{bmatrix} \tag{2.10}$$

By making the last row of $f(\mathbf{x}_s, \mathbf{u}_s)$ (equation 2.8) equal to zero, u_T, s for a given V_s is calculated. It is important to notice that this last row which represents $\dot{V}$ is always equal to zero in steady-state condition. In practice, this linearization is done numerically. Comparing the results of the numerical implementation (`Todo 2.1`) and the analytical derivations above shows the same results. The numerical resulted values for $f(\mathbf{x}_s, \mathbf{u}_s)$, A, and B are the same in both methods, although they are derived in different ways. (The numerical values for the analytical method are calculated by substituting $V_s = 120\frac{km}{h} = 33.3333\frac{m}{s}$ and the resulting $u_{T,s} = 0.2018$ from making the last row of $f(\mathbf{x}_s, \mathbf{u}_s)$ equal to zero, into the analytical expressions derived above.) These numerical values (rounded to four decimal places) are reported below.

$$f(\mathbf{x}_s, \mathbf{u}_s) = \begin{bmatrix} 33.3333 \\ 0 \\ 0 \\ 0 \end{bmatrix}, \ A = \begin{bmatrix} 0 & 0 & 0 & 1 \\ 0 & 0 & 33.3333 & 0 \\ 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & -0.0244 \end{bmatrix}, \ B = \begin{bmatrix} 0 & 0 \\ 20 & 0 \\ 12.8205 & 0 \\ 0 & 1.6667 \end{bmatrix}$$

Here, the derivatives are calculated by hand. The MATLAB file `Deliverable_2_1.m`, included in the codes, performs the same calculations using MATLAB.

2.2. System Decomposition (Deliverable 2.2)

Analyzing matrix A (the Jacobian of the system dynamics f with respect to the state vector) reveals the relationships between states rates of change. For instance, $A(1, 4) = 1$ shows that $\dot{x}$ depends solely on V (and is not affected by any of the lateral states,i.e. y and θ), while $A(2, 3) \neq 0$ indicates that $\dot{y}$ is influenced by θ but not by the longitudinal states x and V. Similarly, $A(4, 4) \leq 0$ shows that $\dot{V}$ is proportional to V, and independent of lateral states.

Matrix B (the Jacobian of f with respect to inputs) illustrates the dependence of states rates of change (and consequently states themselves) on control inputs. For lateral input δ (steering angle), the nonzero entries $B(2, 1)$ and $B(3, 1)$ confirm its influence on lateral states, while longitudinal states remain unaffected. For the longitudinal input u_T (normalized throttle to the motor), $B(4, 2)$ shows its direct effect on $\dot{V}$, and thus an indirect influence on x through V, confirming its influence only on longitudinal states.

This decoupling of longitudinal (related to x and V) and lateral (related to y and θ) dynamics confirms the feasibility of treating the system as two independent subsystems for control design.

- **Longitudinal subsystem**: Responsible for controlling the vehicle's forward (x) position and maintaining the desired velocity. The dynamics can be represented as:

$$\begin{bmatrix} \dot{x} \\ \dot{V} \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \end{bmatrix} + \begin{bmatrix} 0 & 1 \\ 0 & -\frac{U_{T,s} P_{max}}{V_s^2 m} - \frac{\rho C_d A_f V_s}{m} \end{bmatrix} \begin{bmatrix} x - x_s \\ V - V_s \end{bmatrix} + \begin{bmatrix} 0 \\ \frac{P_{max}}{V_s m} \end{bmatrix} (u_T - u_{T,s}) \tag{2.11}$$

- **Lateral subsystem**: Handles the vehicle's heading angle and lateral position within the lane. The dynamics are given by:

$$\begin{bmatrix} \dot{y} \\ \dot{\theta} \end{bmatrix} = \begin{bmatrix} 0 & V_s \\ 0 & 0 \end{bmatrix} \begin{bmatrix} y - y_s \\ \theta - \theta_s \end{bmatrix} + \begin{bmatrix} V_s \frac{\ell_r}{\ell_r + \ell_f} \\ \frac{V_s}{\ell_r + \ell_f} \end{bmatrix} (\delta - \delta_s) \tag{2.12}$$

The following sections provide the detailed controller implementation procedure of these two subsystems.

2.3. MPC Controller Design (Deliverable 3)

2.3.1. Constraints Satisfaction

As shown in equation (1.8) to equation (1.11), there are state and input constraints that must be satisfied. To achieve this, Model Predictive Control (MPC) utilizes several techniques to ensure constraint satisfaction. The cost function for the optimization problem is:

$$\begin{aligned} \min \ & \sum_{i=0}^{N-1} e_i^\top Q e_i + \Delta u_i^\top R \Delta u_i + e_N^\top V_f e_N \\ \text{s.t.} \ & e^+ = A_d e + B_d \Delta u \\ & H_e e_i \leq K_e \\ & e_N \in X_f \end{aligned} \tag{2.13}$$

e_i is the state error at time step i,

Δu_i is the control input change at time step i,

Q, R, V_f are positive definite weighting matrices for the state error, control input change, and terminal state error.,

A_d, B_d are the discrete-time system matrices,

H_e is the constraint matrix for the state error,

K_e is the constraint bound for the state error,

X_f is the terminal set of feasible states.

where e_i and Δu_i are the state error and control input change at time step i, and Q, R, and V_f are positive definite weighting matrices. The terminal constraint ensures $e_N \in X_f$, and the initial state e_0 must lead to a feasible solution.

- **Finite Horizon:** For time steps $i < N$, the optimization problem is solved over a finite prediction horizon N. The optimization at each step minimizes control cost while ensuring the system stays within state and input constraints. The prediction horizon allows the controller to anticipate the system's behavior and adjust accordingly, ensuring the system remains feasible and optimal within the horizon.
- **Terminal Set:** At the terminal step $i = N$, a terminal set X_f is enforced to ensure the system stays within a region where it will continue to satisfy constraints beyond the prediction horizon. This terminal set ensures the system remains feasible even after the optimization ends. The terminal set, together with the finite horizon, guarantees that the system will not violate constraints in the long term.

These two components work together to achieve recursive constraint satisfaction. By solving the optimization problem over a finite horizon, the system is ensured to stay within constraints for the given time steps. Once the terminal set is enforced at the last time step $i = N$, the system is placed in a region where constraints will remain satisfied even beyond the horizon. This combination of finite horizon and terminal set provides recursive feasibility meaning when the optimization is re-solved at the next time step, the new problem will respect the constraints of the previous problem, ensuring that the system continues to satisfy constraints over multiple horizons. The terminal cost aligns with the infinite-horizon optimal control problem using Linear Quadratic Regulator (LQR) theory:

$$J_f(e_N) = e_N^\top V_f e_N$$

where V_f is the terminal cost weighting matrix, ensuring stability and recursive feasibility.

This combination of finite horizon optimization and terminal constraints ensures recursive feasibility in Model Predictive Control (MPC), meaning the system will consistently satisfy constraints and maintain stability, which is essential for the vehicle control problem discussed in this report.

For the longitudinal subsystem, since there are no state constraints, its controller does not require a terminal set. However, for the lateral subsystem, the terminal set is essential to ensure continuous constraint satisfaction. Based on the analysis of linearization and system decomposition, a comparison of these two subsystems is presented in Table 2.1.

Table 2.1: Controller comparison for each subsystems

Comparison Terms	Longitudinal Controller	Lateral Controller
Inputs	u_T	δ
States	V	y, θ
States constraints	–	$-0.5m \leq y \leq 3.5m$ $\lvert\theta\rvert \leq 5° = 0.0873rad$
Input constraints	$-1 \leq u_T \leq 1$	$\lvert\delta\rvert \leq 30° = 0.5236rad$
State transmission equation	$v^+ = av + bu_T$	$e^+ = A_d e + B_d \Delta u$
Terminal set	Not Defined	Configured

2.3.2. Longitudinal Controller

To implement the Model Predictive Control (MPC), discretization is required to allow the state evolution equation to be applied at each prediction time step. The discretized system model for the longitudinal subsystem is given by:

$$\mathbf{x}_{lon}^{+} = f_d(\mathbf{x}_s, u_s) + A_d(\mathbf{x}_{lon} - \mathbf{x}_{lon,s}) + B_d(\mathbf{u}_{lon} - \mathbf{u}_{lon,s}) \tag{2.14}$$

For the longitudinal subsystem, as outlined in table 2.1, there are no state constraints. Consequently, there is no need to design a terminal set for this system. Additionally, when the system reaches the reference velocity, the state x will increase proportionally based on V_{ref}. Since this state doesn't provide significant information and adds unnecessary complexity to the system model, it is omitted for the longitudinal subsystem. The simplified system model is as follows:

$$\Delta V^{+} = A_d \Delta V + B_d \Delta u_T \tag{2.15}$$

This model results in a system with a single input and output, and all values involved in the optimization process are scalar quantities. The setup for the optimization problem is then formulated as:

$$\begin{aligned}
\min \sum_{i=0}^{N-1} & \big((\Delta V_i - \Delta V_{ref})^\top Q(\Delta V_i - \Delta V_{ref}) + (\Delta u_{T,i} - \Delta u_{ref})^\top R(\Delta u_{T,i} - \Delta u_{ref})\big) \\
& + (\Delta V_N - \Delta V_{ref})^\top V_f(\Delta V_N - \Delta V_{ref}) \\
\text{s.t.} \quad & \Delta V^{+} = A_d \Delta V + B_d \Delta u_T \\
& -1 - u_{T,s} \leq \Delta u_{T,i} \leq 1 - u_{T,s}
\end{aligned} \tag{2.16}$$

In equation (2.16), the parameters Q, R, and the prediction horizon H are the key elements that need to be tuned further to enhance the optimization results.

2.3.3. Lateral Controller

Similar to the longitudinal system, before implementing the Model Predictive Control (MPC), we first need to discretize the system models. After discretization, the system model becomes:

Parameter Name	Value
Q	50
R	1
H	5

Table 2.2: Parameters used in the longitudinal controller.

$$\mathbf{x}_{lat}^{+} = f_d(\mathbf{x}_s, u_s) + A_d(\mathbf{x}_{lat} - \mathbf{x}_s) + B_d(\mathbf{u}_{lat} - \mathbf{u}_s) \tag{2.17}$$

For the lateral system, it is evident that $f_s(\mathbf{x}_s, \mathbf{u}_s)$ equals zero. Thus, the model simplifies to:

$$\mathbf{x}_{lat}^{+} = A_d \mathbf{x}_{lat} + B_d \mathbf{u}_{lat} \tag{2.18}$$

To ensure that the vehicle tracks the reference in the lateral direction, we define the state as $\mathbf{e} = \mathbf{x}_{lat} - \mathbf{x}_{ref}$. This gives the new state transition equation:

$$\mathbf{x}_{lat}^{+} - \mathbf{x}_{ref} = A_d(\mathbf{x}_{lat} - \mathbf{x}_{ref}) + B_d(\mathbf{u}_{lat} - \mathbf{u}_{ref}) + (A_d - I)\mathbf{x}_{ref} + B_d \mathbf{u}_{ref} \tag{2.19}$$

Since $\mathbf{x}_{ref}$ and $\mathbf{u}_{ref}$ represent the steady states of the system, we can derive the following relationship:

$$\mathbf{x}_{ref} = A_d \mathbf{x}_{ref} + B_d \mathbf{u}_{ref} \tag{2.20}$$

Substituting equation (2.20) into equation (2.19), we obtain the error dynamic equation:

$$\mathbf{e}^{+} = A_d \mathbf{e} + B_d \Delta \mathbf{u} \tag{2.21}$$

From equation (2.21), it can be seen that for any reference value, the error dynamics remain the same. Therefore, the terminal set can be computed offline. If the reference value changes, the terminal set can easily be adjusted by translating its center to the new reference value.

Regarding the constraints for this system, specific problem characteristics need to be considered. In the context of this vehicle control problem, the car operates on a highway with two lanes. The reference value for the lateral system has two possible cases: either the car should drive in the left lane, where $\mathbf{x}_{ref} = [3, 0]^T$, or in the right lane, where $\mathbf{x}_{ref} = [0, 0]^T$. To define the constraints, two extreme scenarios are considered. When the car is near the boundary of the left lane, the lateral distance between the car's center and the reference position will be 0.6 m, and when near the boundary of the right lane, this distance will be -0.6 m. The reference state for the heading of the car remains zero, and thus, the constraints on the error between the car's heading and the reference value stay the same as the original constraints. Consequently, the new constraints for the error state $\mathbf{e}$ are $[-0.6\,\text{m}, 0.6\,\text{m}]$ for the lateral error $\mathbf{e}_y$ and $[-5°, +5°]$ for the heading error $\mathbf{e}_\theta$.

Based on equation (2.21) and the new constraints, the terminal set is illustrated below:

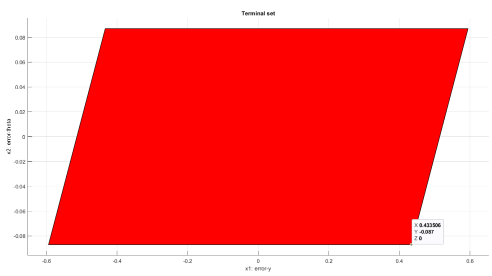


Figure 2.1: Terminal set for lateral subsystem

The optimization problem for the lateral subsystem is formulated as:

$$
\begin{aligned}
\min \sum_{i=0}^{N-1} & \left(\mathbf{e}_i^\top Q \mathbf{e}_i + \Delta\mathbf{u}_i^\top R \Delta\mathbf{u}_i\right) + \mathbf{e}_N^\top V_f \mathbf{e}_N \\
\text{s.t.} \quad & \mathbf{e}^+ = A_d \mathbf{e} + B_d \Delta\mathbf{u} \\
& H_e \mathbf{e}_i \leq K_e \\
& \mathbf{e}_N \in X_f
\end{aligned}
\tag{2.22}
$$

In equation (2.22), the parameters Q, R, and the horizon H are key components that need to be further tuned to optimize the results.

Trade-Off in Tuning: The tuning process balances Q, R, and the horizon N to achieve both trajectory tracking and passenger comfort. Increasing Q improves the system's ability to track the reference trajectory, while increasing R enhances passenger comfort by smoothing control inputs. However, excessively high values of R can lead to unbalanced weights, causing the system to overly prioritize minimizing input changes, which is not reasonable as the control system should not rely too heavily on a single parameter at the expense of overall performance and stability. The prediction horizon H is also critical larger horizons allow the controller to better anticipate future behavior and improve optimization results, but they increase computational cost. Thus, H is tuned to strike a balance between computational efficiency and system performance.

Parameter Name	Value
Q	2
R	20
H	10

(a) Parameters for lateral.

Parameter Name	Value
Q	1
R	2
H	15

(b) Parameters for longitudinal.

2.3.4. Results

figure 2.5 shows the results satisfying constraints and performance requirements for both the lateral and longitudinal controllers.

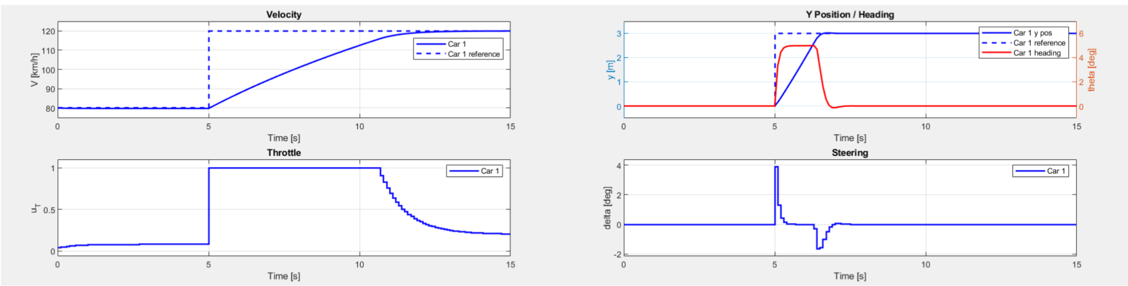


Figure 2.3: Offset free tracking result

2.4. Offset-Free Tracking (Deliverable 4)

Due to linearization errors , tracking velocities other than the linearization point V_s leads to steady-state tracking errors. These errors can be considered to arise through the input u_T as shown in Equation \ref{eq:OffsetFreeGeneral}:

$$
\mathbf{x}^+ = f_d(\mathbf{x}_s, \mathbf{u}_s) + A_d(\mathbf{x} - \mathbf{x}_s) + B_d(\mathbf{u} - \mathbf{u}_s) + \hat{B}_d d \tag{2.23}
$$

Here, d is an unknown constant disturbance, and $\hat{B}_d$is the submatrix of B_d associated with u_T. This section aims to update the longitudinal controller to reject disturbances and eliminate tracking offsets.

2.4.1. Control Modeling

Since the disturbance d only affects the longitudinal subsystem, the state transmission equation in equation (2.24) is derived from the entire system, while the lateral subsystem remains as previously designed.

$$v^+ = v_s + A_d(v - v_s) + B_d(u - u_s) + \hat{B}_d d \tag{2.24}$$

Thus, to achieve offset-free tracking and compensate for this constant error, the model is first augmented by including d as a state, which means that the new state $\mathbf{X}_{new} = [\hat{v}, \hat{d}]^T$. Here, $\hat{v}$ and $\hat{d}$ are estimates of the velocity v and disturbance respectively. Based on the augmented model. The state and disturbance estimator is designed as follows:

$$\begin{bmatrix} \hat{v}_{k+1} \\ \hat{d}_{k+1} \end{bmatrix} = \begin{bmatrix} A_d & B_d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{v}_k \\ \hat{d}_k \end{bmatrix} + \begin{bmatrix} B_d \\ 0 \end{bmatrix} u_k + \begin{bmatrix} L_v \\ L_d \end{bmatrix} (\hat{v}_k - v_k) \tag{2.25}$$

The control goal of the estimator is to let the $\hat{v}$ and $\hat{d}$ converge to the true value.To achieve this, analysing the estimation error synamics is necesssary:

$$\begin{aligned} \begin{bmatrix} v_{k+1} - \hat{v}_{k+1} \\ d_{k+1} - \hat{d}_{k+1} \end{bmatrix} &= \begin{bmatrix} A_d & B_d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} v_k \\ d_k \end{bmatrix} + \begin{bmatrix} B_d \\ 0 \end{bmatrix} u_k \\ &\quad - \begin{bmatrix} A_d & B_d \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \hat{v}_k \\ \hat{d}_k \end{bmatrix} - \begin{bmatrix} B_d \\ 0 \end{bmatrix} u_k - \begin{bmatrix} L_x \\ L_d \end{bmatrix} (\hat{v}_k - v_k) \\ &= \left(\begin{bmatrix} A_d & B_d \\ 0 & 1 \end{bmatrix} + \begin{bmatrix} L_x \\ L_d \end{bmatrix} \begin{bmatrix} 1 & 0 \end{bmatrix} \right) \begin{bmatrix} v_k - \hat{v}_k \\ d_k - \hat{d}_k \end{bmatrix} = A_{bar} \begin{bmatrix} v_k - \hat{v}_k \\ d_k - \hat{d}_k \end{bmatrix} \end{aligned} \tag{2.26}$$

To ensure that the state transmission error ultimately converges to zero, the error dynamics system must be asymptotically stable (AS). Pole placement is an effective method to achieve this by positioning the eigenvalues of the matrix A_{bar} within the unit circle.

2.4.2. Optimizer Design

The optimization problem defined for the offset-free problem is shown below:

$$\begin{aligned} \min \sum_{i=0}^{N-1} & (x_i - x_{ref})^T Q (x_i - x_{ref}) + (u_i - u_{ref})^T R (u_i - u_{ref}) + (x_N - x_{ref})^T V_f (x_N - x_{ref}) \\ s.t. \quad & x_{i+1} = A_d x_i + B_d u_i + d_i \\ & H_u u_i \le K_u \\ & d_0 = 0 \\ & x_0 = V_0 - V_s \end{aligned}$$

From equation (2.27), Q, R horizon H are parameters that need to be further tuned. Moreover, The selection of the poles chosen for the matrix A_bar in equation (2.26) is a crucial step, as it directly impacts the estimation performance. This topic will be further explored in section 2.4.3.

2.4.3. Parameter tuning

Since the disturbance is constant, a highly dynamic estimator is not necessary. If the poles set for the estimator are too small, it will react excessively to estimation errors, potentially leading to instability and oscillatory behavior. So mild poles should be picked and here we use $[0.8, 0.9]$. The total parameters setting on show in table 2.3.

Parameter Name	Value
Q	10
R	1
poles	[0.8,0.9]

Table 2.3: Parameters used in the offset free controller.

2.4.4. Results

The result is shown in figure 2.4. To observe the effect of the estimator, figure 2.5 compare the steady state error when there is an estimator and not. It can be observed that when there is no estimator, just like what has been implemented previously, the steady state error remains 0.18. After the estimator has been added, the steady state error decreased to 0.0014. This shows that the estimator can somehow deduce the steady state error when we set the same reference velocity. The reason why there still exists an error is because the model only use proportional term to estimate the state and disturbance. This error could be canceled by introducing the integrator which is not being discussed here.

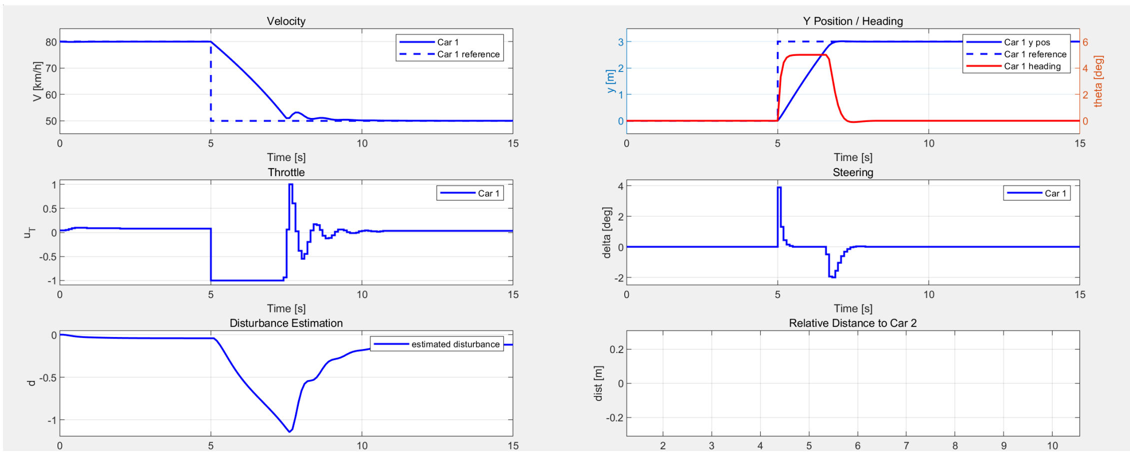
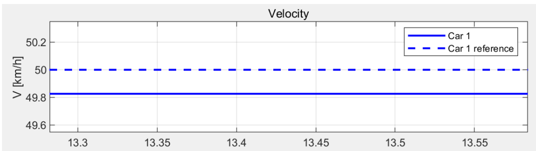
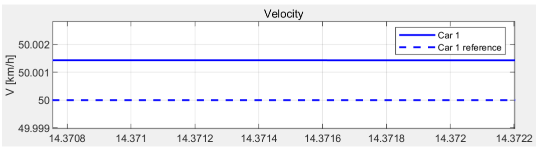


Figure 2.4: Offset free tracking result



(a) Steady state error without estimator



(b) Steady state error with estimator

Figure 2.5: The effect of estimator in reducing the steady state error

2.5. Robust Tube MPC (Deliverable 5)

2.5.1. Control Modeling

In this section, an additional car is introduced to the system. To distinguish between the two vehicles, the car that can be directly controlled is referred to as the 'ego' car, while the car ahead of the ego car is called the 'lead' car. The control objectives are divided into two key aspects:

- **Remaining Distance**:The distance between the 'ego' car and the 'lead' car should be maintained at a specific value, denoted as x_{safe}.
- **Distance constraints**:The distance between the 'ego' car and the 'lead' car should never fall below $6m$ during the tracking process.

The following analysis is based on the assumption that the 'lead' car and the 'ego' car have similar dynamic models, as shown in equation (2.27) and equation (2.28), respectively.

$$\tilde{\mathbf{x}}^{+} = f_d(\mathbf{x}_s, \mathbf{u}_s) + A_d(\tilde{\mathbf{x}} - \mathbf{x}_s) + B_d(\tilde{\mathbf{u}} - \mathbf{u}_s) \tag{2.27}$$

$$\mathbf{x}^{+} = f_d(\mathbf{x}_s, \mathbf{u}_s) + A_d(\mathbf{x} - \mathbf{x}_s) + B_d(\mathbf{u} - \mathbf{u}_s) \tag{2.28}$$

As mentioned previously, the control objective in this section is for the 'ego' car to track the position x of the 'lead' car while maintaining a minimum distance from the car ahead. To achieve this, only the longitudinal direction needs to be considered, while the lateral direction can remain consistent with the previous steps in section 2.3. By subtracting equation (2.28) from equation (2.27) and focusing solely on the longitudinal direction, the following equation is obtained:

$$\tilde{\mathbf{x}}^{+} - \mathbf{x}^{+} = A_d(\tilde{\mathbf{x}} - \mathbf{x}) + B_d(\tilde{u}_T - u_T) \tag{2.29}$$

To control the relative distance between the 'ego' car and the 'lead' car, a new state is defined as $\mathbf{\Delta} = [\Delta_x; \Delta_V] = \tilde{\mathbf{x}}_{long} - \mathbf{x}_{long} - \mathbf{x}_{safe}$, where $\mathbf{x}_{safe} = [x_{safe}; 0]$. x_{safe} here represents the anticipated distance between two cars when the system reaches a steady state. Based on this, the equation (2.29) can be rewritten as follows:

$$\tilde{\mathbf{x}}^{+} - \mathbf{x}^{+} - \mathbf{x}_{safe} = A_d(\tilde{\mathbf{x}} - \mathbf{x} - \mathbf{x}_{safe}) - B_d(u_T - u_s) + B_d(\tilde{u}_T - u_S) + A_d\mathbf{x}_{safe} - \mathbf{x}_{safe} \tag{2.30}$$

Where $A_d = [1, 0.0999; 0, 0.9976]$, such that $A_d\mathbf{x}_{safe}$ exactly yields $\mathbf{x}_{safe}$, therefore equation (2.30) can be simplified as follows:

$$\mathbf{\Delta}^{+} = A_d\mathbf{\Delta} - B_d\Delta u_T + B_d\Delta\tilde{u}_T \tag{2.31}$$

Here, the $\Delta\tilde{u}_T$ can be treated as a time-varying disturbance to the system, with a range of $[-0.5, 0.5]$

2.5.2. Optimizer Design

Then, based on the tube MPC model, we can treat the system as a combination of two subsystems.

- **Noise-free system:**In this system, we don't care about the nosie and just focus on the state transmission as if there exists no noise at all, this system can be defined as the equation below:

$$z_{i+1} = Az_i + Bv_i \tag{2.32}$$

- **Error compensation:**A portion of input to compensate about the error that is introduced into our systems. To achieve this, we need to first design the K controller.;

The optimizer setup for the tube MPC controller is defined as below:

$$\begin{aligned}
\min \sum_{i=0}^{N-1} & (z_i - z_{ref})^T Q (z_i - z_{ref}) + (v_i - v_{ref})^T R (v_i - v_{ref}) + (z_N - z_{ref})^T V_f (z_N - z_{ref}) \\
s.t. \quad & z_{i+1} = A_d z_i + B_d v_i \\
& z_N \in X_f \\
& v_i \in U - K\epsilon \\
& z_i \in X - \epsilon \\
& x_0 = \Delta_0 + \epsilon
\end{aligned}$$

And then, the input that is really used to apply to our system is derived in the equation below:

$$u_{opt} = K(x - z_0^*) + v_0^* \tag{2.33}$$

2.5.3. Parameters Tuning

Key parameters for the Tube MPC controller are summarized in Table 2.4:

Parameter	Value
Q	50
R	1
x_{safe}	10

Table 2.4: Parameters used in the tube MPC controller.

(1) Tuning of safe distance

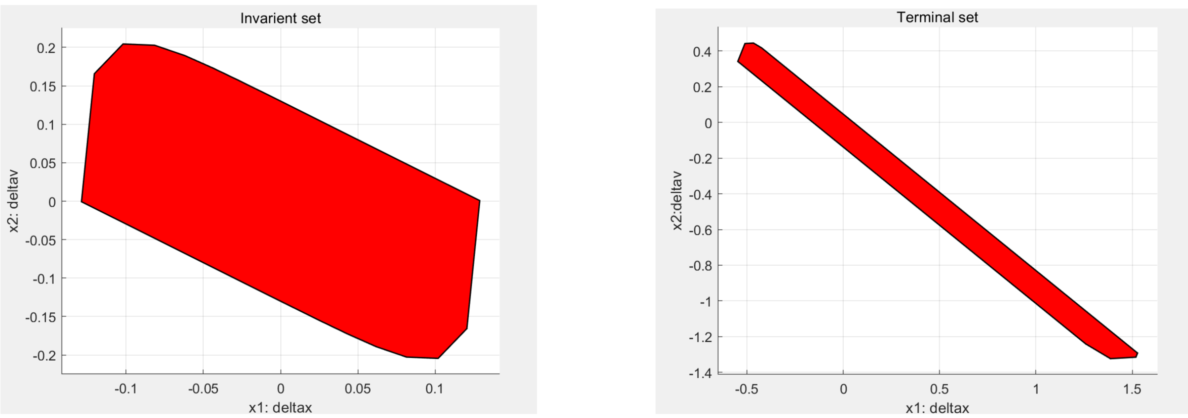
The parameter X_{safe} represents the desired steady steady-distance between the ego car and the lead car under normal driving conditions. It is essential to highlight the constraints applied to the controller. Specifically, the distance between the ego car and the vehicle in front must remain greater than 6 meters, as outlined in the project description. Based on this requirement, the following constraint is derived:

$$z_i = \tilde{V}_i - V_i - x_{safe} \leq [6; 0] - x_{safe} \tag{2.34}$$

To ensure that the steady state remains within the defined state constraints, s_{safe} must be greater than 6. If it is too small, the steady state will be close to the constraints boundary which could introduce instability to the system; if it is too big, than the tracking performance will decrease. In this case, it is set to 10.

(2) Tunning of LQR controller

Tuning the parameters Q and R involves making a trade-off between tracking performance and system stability.
If Q is set much larger than R, the optimizer will prioritize minimizing tracking errors, even if this results in a small input weight relative to the nominal control problem. This can lead to a very small terminal set X_f, which may improve the stability of the system.
In contrast, if R is larger than Q, the optimizer will place more emphasis on the input, causing the tube MPC controller to allocate less input to compensate for dynamic disturbances. This leads to a smaller invariant set and a larger terminal set, which enhances system stability at the cost of reduced tracking performance.
Here, we choose a controller that provides good tracking performance, though at the cost of some stability, as shown on the table 2.4. Based on the selected parameters, the corresponding invariant set ε and the terminal set X_f are illustrated in figure 2.6a and figure 2.6b respectively.



(a) Invariant set ε **(b)** Terminal set

Figure 2.6: Robust tube MPC components

2.5.4. Results

The simulation results are all based on the parameter in table 2.4. For the constant disturbance, the result is shown as below: For the time varying distance, the result is shown as below: From the results, it can be observed that the tube MPC controller effectively handles the constant disturbance, similar to the estimator discussed in section 2.4. Moreover, it responds well to time-varying disturbances, demonstrating its strong dynamic capabilities.

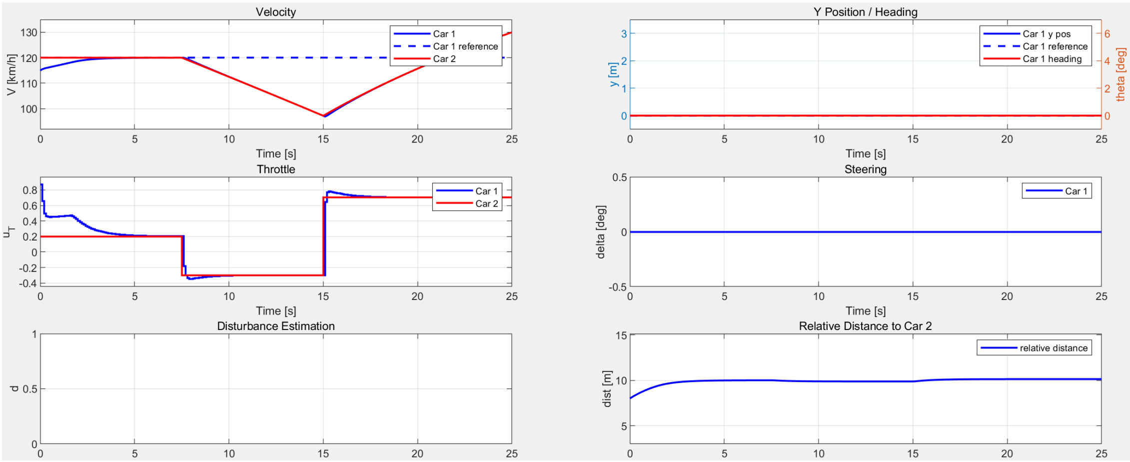


Figure 2.7: Robust tube MPC result

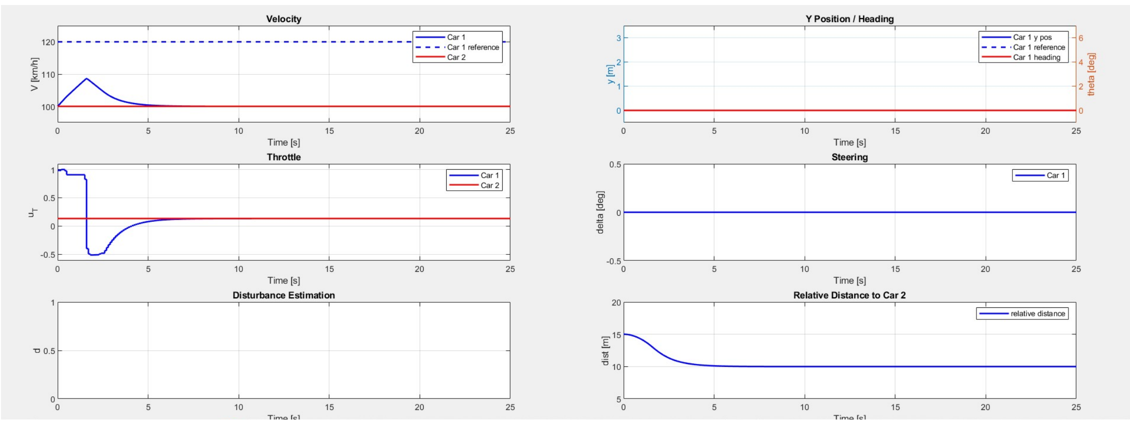


Figure 2.8: Robust tube MPC result

3

Nonlinear MPC (Deliverable 6)

3.1. Tracking Nonlinear MPC Controller (Deliverable 6.1)

The Nonlinear Model Predictive Control (NMPC) controller is designed to:

1. **Reference Tracking:** Accurately track a given reference trajectory for lateral position (y) and velocity (V).
2. **Constraint Satisfaction:** Ensure compliance with operational constraints at all times.
3. **Performance Requirement:** Achieve a settling time of no more than 5 seconds when accelerating from $80\,\mathrm{km/h}$ to $100\,\mathrm{km/h}$ while completing a lane change of $3\,\mathrm{m}$ width.

3.1.1. Discretization of Dynamics

To implement NMPC, the continuous-time dynamics are discretized using the fourth-order Runge-Kutta (RK4) method with a time step T_s. The discrete dynamics for the next state are computed as:

$$\mathbf{x}_{k+1} = f_d(\mathbf{x}_k, \mathbf{u}_k) = \mathbf{x}_k + \frac{T_s}{6}\left(\mathbf{k}_1 + 2\mathbf{k}_2 + 2\mathbf{k}_3 + \mathbf{k}_4\right),$$

where:

$$\begin{aligned}
\mathbf{k}_1 &= f(\mathbf{x}_k, \mathbf{u}_k), \\
\mathbf{k}_2 &= f\left(\mathbf{x}_k + \frac{T_s}{2}\mathbf{k}_1, \mathbf{u}_k\right), \\
\mathbf{k}_3 &= f\left(\mathbf{x}_k + \frac{T_s}{2}\mathbf{k}_2, \mathbf{u}_k\right), \\
\mathbf{k}_4 &= f\left(\mathbf{x}_k + T_s\mathbf{k}_3, \mathbf{u}_k\right).
\end{aligned}$$

This method ensures accurate state prediction over the prediction horizon and maintains numerical stability.

3.1.2. NMPC Cost Function

The NMPC cost function minimizes deviation from the reference trajectory and penalizes control effort:

$$\min \sum_{k=1}^{N} \left(w_1\delta(k)^2 + w_2u_T(k)^2 + w_3(y(k) - y_{\text{ref}})^2 + w_4(V(k) - V_{\text{ref}})^2 + w_5\mathbf{X}(3,k)^2\right)$$

s.t. (3.1)

$$g(\mathbf{x}_i, \mathbf{u}_i) \leq 0, \tag{3.2}$$

$$\mathbf{x}_{i+1} = f_d(\mathbf{x}_i, \mathbf{u}_i).$$

The weights w_1 to w_5 prioritize the following:

- w_1: Penalty for steering angle deviation.
- w_2: Penalty for control effort $u_T(k)$.
- w_3: Deviation in lateral position $y(k)$.
- w_4: Deviation in velocity $V(k)$.
- w_5: Smoothing of heading angle $\mathbf{X}(3, k)$.

The optimization is implemented in CasADi, where the decision variables include states $\mathbf{x}_k$ and controls $\mathbf{u}_k$ over the prediction horizon.

3.1.3. Updated Cost Function for Lane Change Maneuver

For lane change maneuvers, the cost function focuses on smooth control of steering and throttle and maintaining vehicle position along the desired path. The updated weights are:

1. $w_1 = 10000$: Penalizes excessive steering angle control inputs for smoother lane change.
2. $w_2 = 1$: Penalizes sharp throttle acceleration or deceleration.
3. $w_3 = 400$: Penalizes deviations in heading for smooth vehicle turns.
4. $w_4 = 2$: Penalizes lateral position deviation from the reference.
5. $w_5 = 1$: Penalizes longitudinal position deviation from the reference.

3.1.4. Results for Lane Change

The NMPC controller successfully tracks the references while satisfying constraints and performance requirements. The following performance was achieved:

- Settling time of no more than 5 seconds when accelerating from 80 km/h to 100 km/h.
- Successful execution of the lane change maneuver with a lane width of 3 m.

The results from the simulation demonstrate that the NMPC controller successfully executes the lane change maneuver while meeting the required performance criteria. The vehicle accelerates from 80 km/h to 100 km/h smoothly, and the lane change is completed within the targeted time.

The steering input, δ, exhibits smooth transitions, ensuring a comfortable lane change. Similarly, the throttle input is also smooth, with little abrupt changes, contributing to a seamless driving experience. The vehicle follows the reference trajectory closely, achieving the desired position and velocity within the specified time.

The heading remains within the defined constraints throughout the maneuver, maintaining stability. The controller effectively satisfies the lane change requirements while ensuring the vehicle stays on course, with both the steering and throttle inputs behaving in a controlled and smooth manner.

As shown in Figure 3.1, the controller performs the lane change efficiently while maintaining trajectory tracking and constraint satisfaction.

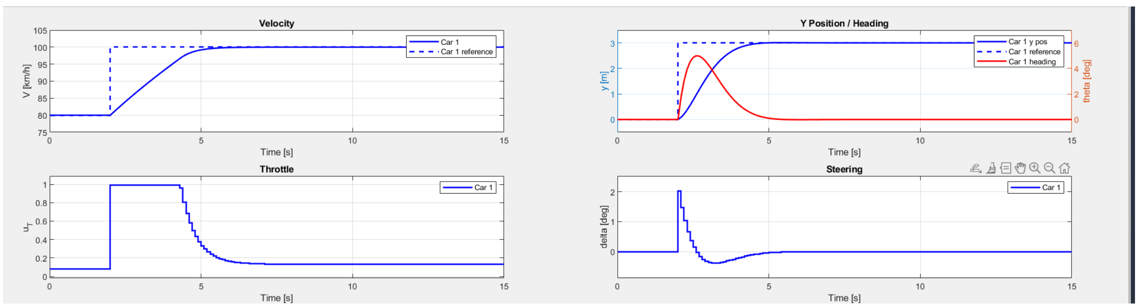


Figure 3.1: Tracking NMPC controller result.

3.1.5. Steady State tracking error

The designed NMPC controller does not result in any steady-state tracking error because it directly uses the full nonlinear system model without any approximations and linearizations. It can accurately track any desired reference, provided it complies with the relevant constraints. By inherently incorporating the exact system dynamics, it avoids errors introduced by linearizing the dynamics around the steady-state condition, such as those caused by motor and drag effects. NMPC explicitly optimizes the system's behavior over the prediction horizon, ensuring accurate tracking of the desired references.

The resulting simulation plots validate this claim. In this problem, the steady state is defined as satisfying the condition in Equation 3.3:

$$f_s(x_s, u_s) = \begin{bmatrix} \dot{x} \\ \dot{y} \\ \dot{\theta} \\ \dot{V} \end{bmatrix} = \begin{bmatrix} V_s \\ 0 \\ 0 \\ 0 \end{bmatrix}, \tag{3.3}$$

The plots also demonstrate that for any arbitrary reference velocity V_s, the car reaches V_s at its steady-state condition, confirming that $\dot{x} = V_s$. Additionally, the plots show that the car maintains a constant velocity at V_s, proving that $\dot{V} = 0$, which indicates no acceleration. Similarly, the y-position and the car's heading angle θ remain constant after reaching the steady-state condition, verifying that $\dot{y} = 0$ and $\dot{\theta} = 0$. (Even θ itself equals 0 at the steady state.)

3.2. MPC Controller for Overtaking (Deliverable 6.2)

This task involved designing an NMPC controller for overtaking a vehicle on the road. The state of the other car is dynamically updated during the prediction horizon to ensure realistic collision avoidance.

3.2.1. Implementation and Validation with Dynamic Vehicle State Updates

The position of the other car along the x-axis (x_{car}) is updated at each time step as:

$$x_{\text{car},k+1} = x_{\text{car},k} + v_{\text{car}} \cdot T_s,$$

where:

- $x_{\text{car},k}$ is the current position of the other car.
- v_{car} is the constant velocity of the other car.
- T_s is the time step.

3.2.2. Collision Avoidance Constraint

An ellipsoidal constraint ensures collision avoidance with the vehicle ahead:

$$\Delta\mathbf{x}^T H \Delta\mathbf{x} \geq 1,$$

where $H = \text{diag}\left(\frac{1}{(L+3)^2}, \frac{1}{(W+1)^2}\right)$, with $L = 3\,\text{m}$ and $W = 1.8\,\text{m}$.

3.2.3. Updated Cost Function for Overtaking Maneuver

For the overtaking maneuver, the cost function is adjusted with the following updated weights:

1. $w_1 = 7000$: Penalizes excessive steering angle control inputs during overtaking.
2. $w_2 = 11$: Penalizes sharp throttle acceleration or deceleration.
3. $w_3 = 3$: Penalizes deviations in heading for smooth vehicle turns.
4. $w_4 = 0.5$: Penalizes lateral position deviation during overtaking.
5. $w_5 = 3$: Penalizes longitudinal position deviation during overtaking.

3.2.4. Results for Overtaking

The NMPC controller performs the overtaking maneuver efficiently, adhering to the desired performance criteria. The vehicle accelerates smoothly from 80 km/h to 120 km/h, completing the overtaking process within the targeted time frame.

The steering input is smooth and controlled, ensuring a stable and comfortable maneuver while overtaking. Similarly, the throttle input shows no abrupt changes, providing a steady acceleration that maintains comfort during the overtaking process. The vehicle closely follows the reference trajectory, achieving the intended position and velocity within the specified time.

Throughout the maneuver, the heading remains within the predefined constraints, ensuring that the vehicle maintains a stable trajectory. The controller successfully satisfies the overtaking objectives while ensuring smooth and controlled inputs for both steering and throttle, demonstrating effective execution of the overtaking strategy.

As shown in Figure 3.2, the controller performs the overtaking maneuver with dynamic updates for the other car.

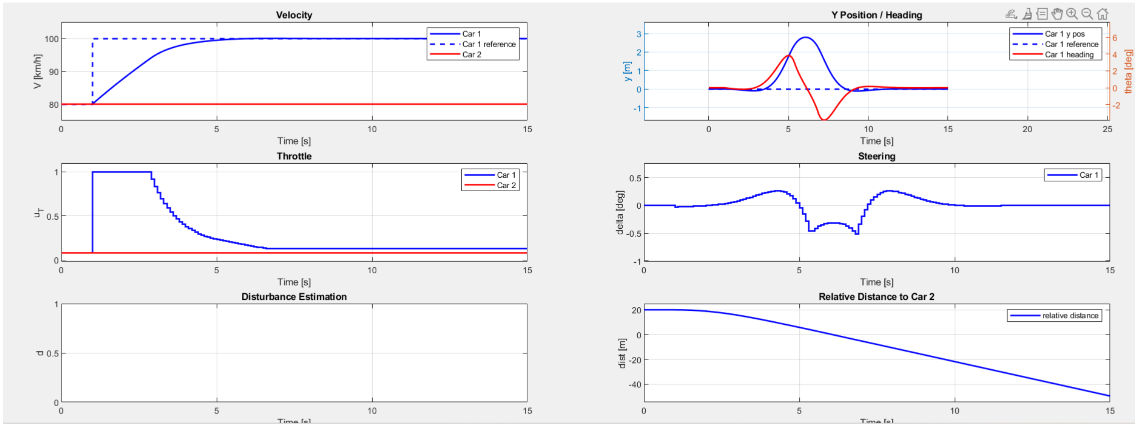


Figure 3.2: NMPC controller result during overtaking with dynamic updates for the other car.